

Year 7 • Chapter 9 • The number plane: answers (teacher copy)

9.01 Grid references on maps (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: the lake
2. Ex 1b: the library
3. Ex 2a: D9
4. Ex 2b: B2, C2, B3, C3
5. Ex 3a: e.g. walk to Gauss Rd, west to Euclid St, then north
6. Ex 3b: it is a convention so everyone reads it the same way

Easy • Set A

1. lake
2. library
3. stadium
4. student centre

Easy • Set A2 (quick drill)

1. car park
2. car park
3. lake
4. stadium
5. student centre
6. main campus

Easy • Set B

1. D9
2. F2 or G2
3. A8 or B8
4. B2, B3, C2 or C3
5. H7, H8, I7 or I8
6. G10, H10 or I10

Medium • Set C

1. e.g. A6
2. e.g. E1
3. E6
4. main campus (row 10)
5. 4 squares
6. student centre and main campus

Challenging • Set D

1. own map
2. own directions

Skill drill 1 (Easy basics)

- | | | |
|----------|----------|--------|
| 1. Cafe | 9. Farm | 17. A1 |
| 2. Fire | 10. Bank | 18. H2 |
| 3. Park | 11. Bus | 19. B2 |
| 4. Wharf | 12. Oval | 20. F6 |
| 5. Gym | 13. E5 | 21. H5 |
| 6. Post | 14. G3 | 22. H6 |
| 7. Zoo | 15. E2 | 23. B6 |
| 8. Pool | 16. F1 | 24. C4 |

Skill drill 2 (Medium basics)

- | | | |
|-------|---------------------|---------------------|
| 1. G6 | 8. H6 (Hotel) | 15. 6 left, 1 up |
| 2. A5 | 9. B2 (Shop) | 16. 0 right, 4 down |
| 3. D4 | 10. E5 (Club) | 17. 5 right, 1 up |
| 4. F3 | 11. 2 right, 0 up | 18. 2 right, 0 up |
| 5. D4 | 12. 0 right, 3 down | 19. 1 left, 2 down |
| 6. H1 | 13. 5 right, 2 up | 20. 5 left, 5 down |
| 7. C5 | 14. 1 left, 1 up | |

Extension

1. C5
2. 18 squares

Exit ticket

1. Level 1: stadium
2. Level 2: F2
3. Level 3: student centre (H7)

9.02 Points in the first quadrant (booklet p. 11)

11)

WE DO examples (teacher)

1. Ex 1a: plotted
2. Ex 1b: plotted on the axes
3. Ex 2a: P(4, 1), Q(2, 5)
4. Ex 2b: No: (2, 5) is 2 across, 5 up
5. Ex 3a: rectangle
6. Ex 3b: (0, 0)

Easy • Set A

1. plotted
2. plotted

Easy • Set A2 (quick drill)

1. 4
2. 7
3. 0
4. 0
5. 9
6. 8

Easy • Set B

1. (1, 2)
2. (4, 5)
3. (7, 1)
4. (0, 3)
5. (5, 0)
6. (6, 4)

Medium • Set C

1. rectangle
2. isosceles triangle

Challenging • Set D

1. (1, 6)
2. x-axis points have $y = 0$; y-axis points have $x = 0$

Skill drill 1 (Easy basics)

- | | | |
|------------|------------|--------|
| 1. (7, 8) | 11. (1, 2) | 21. 8 |
| 2. (0, 0) | 12. (0, 3) | 22. 4 |
| 3. (4, 6) | 13. 6 | 23. 10 |
| 4. (4, 5) | 14. 10 | 24. 0 |
| 5. (2, 6) | 15. 4 | 25. 0 |
| 6. (6, 6) | 16. 1 | 26. 3 |
| 7. (1, 8) | 17. 12 | 27. 7 |
| 8. (6, 0) | 18. 12 | 28. 10 |
| 9. (8, 8) | 19. 6 | |
| 10. (2, 8) | 20. 7 | |

Skill drill 2 (Medium basics)

- | | | |
|------|-------|-------|
| 1. B | 7. D | 13. 4 |
| 2. J | 8. H | 14. 3 |
| 3. M | 9. K | 15. 8 |
| 4. F | 10. L | 16. 3 |
| 5. C | 11. A | 17. 5 |
| 6. E | 12. G | 18. 2 |

- | | | |
|-------|-------|-------|
| 19. 1 | 21. 2 | 23. 3 |
| 20. 3 | 22. 2 | 24. 3 |

Extension

- (3, 5)
- 24 square units

Exit ticket

- Level 1: 7
- Level 2: on the y-axis, 5 up
- Level 3: (1, 5)

9.03 The number plane (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: 1st top right, 2nd top left, 3rd bottom left, 4th bottom right
- Ex 1b: 3rd
- Ex 2a: A(2, 3), B(-3, 1), C(-2, -4), D(4, -2)
- Ex 2b: 4th
- Ex 3a: a zigzag line
- Ex 3b: e.g. (-2, 3); e.g. (0, 5)

Easy · Set A (quick drill)

- | | | |
|-----------|------------|------------|
| 1. 3rd | 6. x-axis | 11. 1st |
| 2. 1st | 7. 4th | 12. x-axis |
| 3. 2nd | 8. 3rd | 13. 4th |
| 4. 4th | 9. 2nd | 14. 2nd |
| 5. y-axis | 10. y-axis | 15. origin |

Easy · Set B

- (1, 3)
- (-4, 2)
- (-3, -3)
- (3, -2)
- (0, -4)
- (-5, 0)

Medium · Set C

- a flower / star shape (symmetrical)
- D(-4, 3)

Challenging · Set D

- B: 2nd
- (0, 0)
- (-3, -2)
- 7 units

Skill drill 1 (Easy basics)

- | | | |
|---------|------------|--------------|
| 1. 3rd | 13. 1st | 25. (4, 3) |
| 2. 2nd | 14. 2nd | 26. (3, -4) |
| 3. 2nd | 15. 1st | 27. (-5, -1) |
| 4. 4th | 16. 4th | 28. (-1, -5) |
| 5. 4th | 17. 2nd | 29. (-1, -1) |
| 6. 3rd | 18. 4th | 30. (5, -2) |
| 7. 3rd | 19. 1st | 31. (-1, -2) |
| 8. 2nd | 20. 3rd | 32. (3, -1) |
| 9. 2nd | 21. 2nd | 33. (-1, -3) |
| 10. 4th | 22. 3rd | 34. (4, 4) |
| 11. 2nd | 23. 2nd | 35. (0, -2) |
| 12. 3rd | 24. x-axis | 36. (-4, 3) |

Skill drill 2 (Medium basics)

- | | | |
|------------|-------------|-------------------|
| 1. (7, -7) | 7. (-7, -3) | 13. (6, -8) |
| 2. (3, 3) | 8. (-4, 4) | 14. (-2, -4) |
| 3. (-4, 2) | 9. (4, -8) | 15. (-1, 8) |
| 4. (-4, 1) | 10. (2, -6) | 16. (-5, 2) |
| 5. (-1, 4) | 11. (8, 7) | 17. 1st |
| 6. (4, 6) | 12. (5, 6) | 18. on the y-axis |

- | | | |
|---------|-------------------|----------------|
| 19. 1st | 23. 2nd | 27. 3rd |
| 20. 2nd | 24. on the x-axis | 28. the origin |
| 21. 4th | 25. 3rd | |
| 22. 4th | 26. 2nd | |

Extension

- 12 square units
- (1, -1), (4, -1), (1, -3)

Exit ticket

- Level 1: 2nd
- Level 2: e.g. (3, -1)
- Level 3: (-2, -5)

9.04 Plotting tables of values (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: vertical line $x = -2$
- Ex 1b: horizontal line $y = 3$
- Ex 2a: straight line going down
- Ex 2b: straight line going up steeply
- Ex 3a: y goes up by 2
- Ex 3b: $y = 7$

Easy · Set A

- vertical line
- horizontal line

Easy · Set A2 (quick drill)

- (0, 3)
- (2, -1)
- (-4, 0)
- (-1, -5)
- (5, 5)
- (0, 0)

Easy · Set B

- (0, 5), (1, 4), (2, 3)
- (-1, -1), (0, 1), (1, 3)
- (1, 2), (2, 4), (3, 6)
- (0, 1), (2, 1), (4, 1)

Medium · Set C

- straight line going down
- straight line going up

Challenging · Set D

- a U-shaped curve (parabola)
- no, it is a curve

Skill drill 1 (Easy basics)

- | | | |
|-------------------------------------|--------------------------------------|-------------|
| 1. (0, 5), (1, 9), (2, 13), (3, 17) | 7. (0, 6), (1, 10), (2, 14), (3, 18) | 15. (3, 16) |
| 2. (0, 0), (1, 3), (2, 6), (3, 9) | 8. (0, 3), (1, 6), (2, 9), (3, 12) | 16. (3, 5) |
| 3. (0, 5), (1, 7), (2, 9), (3, 11) | 9. (0, 1), (1, 4), (2, 7), (3, 10) | 17. (3, 15) |
| 4. (0, 2), (1, 3), (2, 4), (3, 5) | 10. (0, 5), (1, 6), (2, 7), (3, 8) | 18. (4, 17) |
| 5. (0, 1), (1, 3), (2, 5), (3, 7) | 11. (6, 25) | 19. (6, 15) |
| 6. (0, 3), (1, 7), (2, 11), (3, 15) | 12. (4, 16) | 20. (4, 12) |
| | 13. (4, 4) | 21. (3, 14) |
| | 14. (6, 30) | 22. (6, 9) |
| | | 23. (3, 10) |
| | | 24. (4, 15) |
| | | 25. (6, 28) |

Skill drill 2 (Medium basics)

- | | | |
|---------------------|---------------------|--------------------|
| 1. y goes down by 1 | 4. y goes down by 2 | 7. y goes up by 3 |
| 2. y goes up by 3 | 5. y goes down by 3 | 8. y goes up by 1 |
| 3. y goes up by 2 | 6. y goes up by 2 | 9. y goes up by 4 |
| | | 10. y goes up by 5 |

11. y goes down by 3
12. y goes up by 4
13. 24
14. 13
15. 15
16. 19
17. 29
18. 28
19. 3
20. 24
21. 25

Extension

1. -5
2. $y = 4x$

Exit ticket

1. Level 1: (0, 3), (1, 5), (2, 7)
2. Level 2: horizontal line
3. Level 3: it goes up by 3

9.05 Tables of values from rules (booklet p.

35)

WE DO examples (teacher)

1. Ex 1a: 2, 3, 4, 5, 6
2. Ex 1b: -11, -9, -7, -5, -3
3. Ex 2a: 6, 5, 4, 3, 2
4. Ex 2b: -5, -2, 1, 4, 7
5. Ex 3a: $y = 2x + 4$
6. Ex 3b: $y = 8 - x$

Easy · Set A

1. 2, 3, 4, 5, 6
2. -2, 0, 2, 4, 6
3. -4, -3, -2, -1, 0
4. -10, -5, 0, 5, 10

Easy · Set A2 (quick drill)

1. 6
2. 15
3. -6
4. 7
5. 6
6. -2
7. 3
8. -7
9. 4

Easy · Set B

1. -11, -9, -7, -5, -3
2. 6, 5, 4, 3, 2

Medium · Set C

1. -5, -2, 1, 4, 7
2. 7, 5, 3, 1, -1
3. -4, -3, -2, -1, 0
4. 10, 7, 4, 1, -2

Challenging · Set D

1. $y = 2x + 4$
2. $y = 8 - x$
3. $y = 4x - 3$
4. $y = -3x$

Skill drill 1 (Easy basics)

- | | | |
|-------|--------|--------|
| 1. 15 | 9. 19 | 17. 10 |
| 2. 14 | 10. 15 | 18. 18 |
| 3. 16 | 11. -1 | 19. 37 |
| 4. 6 | 12. 15 | 20. 4 |
| 5. 22 | 13. 5 | 21. 20 |
| 6. 2 | 14. 21 | 22. 26 |
| 7. 5 | 15. 14 | 23. 4 |
| 8. 5 | 16. 45 | 24. 12 |

25. 8, 11, 14, 17

26. 4, 8, 12, 16

27. 8, 9, 10, 11

28. 9, 12, 15, 18

29. 3, 7, 11, 15

30. 6, 7, 8, 9

31. 6, 10, 14, 18

32. 0, 5, 10, 15

33. 3, 5, 7, 9

34. 9, 10, 11, 12

Skill drill 2 (Medium basics)

- | | | |
|-----------------|------------------|--------------|
| 1. $y = x + 1$ | 10. $y = x + 7$ | 19. $x = 4$ |
| 2. $y = 5x + 6$ | 11. $y = 6x + 6$ | 20. $x = 1$ |
| 3. $y = x + 2$ | 12. $y = 4x + 1$ | 21. $x = 0$ |
| 4. $y = 4x$ | 13. $x = 2$ | 22. $x = 6$ |
| 5. $y = x + 4$ | 14. $x = 9$ | 23. $x = 6$ |
| 6. $y = 2x + 4$ | 15. $x = 8$ | 24. $x = 7$ |
| 7. $y = 6x$ | 16. $x = 1$ | 25. $x = 10$ |
| 8. $y = 3x$ | 17. $x = 5$ | 26. $x = 0$ |
| 9. $y = 2x + 2$ | 18. $x = 8$ | 27. $x = 9$ |

Extension

1. 2, 0, 0, 2, 6
2. $y = x^2$

Exit ticket

1. Level 1: 3
2. Level 2: -1, 3, 7
3. Level 3: $y = 3x + 1$

9.06 Graphing linear equations (booklet p. 43)

WE DO examples (teacher)

1. Ex 1a: $y: -3, -1, 1, 3$; straight line
2. Ex 1b: line through (0, 3) going down
3. Ex 2a: yes
4. Ex 2b: yes
5. Ex 3a: $y: -4, -3, -2, -1, 0$
6. Ex 3b: (0, -1)

Easy · Set A

1. $y: -1, 0, 1, 2, 3$
2. $y: -3, -1, 1, 3$

Easy · Set A2 (quick drill)

1. yes
2. yes
3. no
4. yes
5. yes
6. no

Easy · Set B

1. no
2. no
3. yes
4. yes
5. yes
6. yes

Medium · Set C

1. $y: 5, 4, 3, 2, 1$
2. $y: -4, -3, -2, -1, 0$

Challenging · Set D

1. parallel lines, 2 units apart
2. (1, 2)

Skill drill 1 (Easy basics)

- | | | |
|--------|---------|---------|
| 1. yes | 7. yes | 13. no |
| 2. yes | 8. no | 14. yes |
| 3. yes | 9. yes | 15. yes |
| 4. no | 10. no | 16. no |
| 5. yes | 11. no | 17. yes |
| 6. yes | 12. yes | 18. yes |

- | | | | | | |
|---------|----------------------------|-------------------------|---------------|---------------|---------------|
| 19. no | 25. $(-1, -3), (0, -1),$ | 29. $(-1, -2), (0, 1),$ | 13. 9 | 19. $(0, -2)$ | 25. $(0, 9)$ |
| 20. yes | $(1, 1), (2, 3)$ | $(1, 4), (2, 7)$ | 14. 19 | 20. $(0, -5)$ | 26. $(0, -6)$ |
| 21. yes | 26. $(-1, -4), (0, -1),$ | 30. $(-1, -1), (0, 0),$ | 15. -1 | 21. $(0, 7)$ | 27. $(0, 0)$ |
| 22. no | $(1, 2), (2, 5)$ | $(1, 1), (2, 2)$ | 16. 10 | 22. $(0, 6)$ | 28. $(0, -3)$ |
| 23. no | 27. $(-1, 2), (0, 3), (1,$ | 31. $(-1, -3), (0, 0),$ | 17. $(0, -3)$ | 23. $(0, 9)$ | |
| 24. yes | $4), (2, 5)$ | $(1, 3), (2, 6)$ | 18. $(0, -1)$ | 24. $(0, 3)$ | |
| | 28. $(-1, 0), (0, 2), (1,$ | 32. $(-1, -1), (0, 2),$ | | | |
| | $4), (2, 6)$ | $(1, 5), (2, 8)$ | | | |

Skill drill 2 (Medium basics)

- | | | |
|---------|---------|-----------|
| 1. -4 | 5. -6 | 9. -4 |
| 2. 3 | 6. 0 | 10. -10 |
| 3. 0 | 7. 11 | 11. -1 |
| 4. 4 | 8. 2 | 12. 12 |

Extension

- $m = 3$
- $y = 2x + 1$ and $y = 2x - 3$

Exit ticket

- Level 1: 5
- Level 2: yes
- Level 3: yes